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Hydrodynamics of Confined Swimming: From Stokes flow to intermediate inertial regime

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We investigate the dynamics of a swimmer in a confined environment. Our findings show that the transition from the Stokes regime to the intermediate intertial regime can be tuned solely by adjusting the swimmer's confinement. The crossover between the two is governed by the formation of a boundary layer and occurs when the confinement length scale becomes larger than the Blasius boundary layer thickness. We further support these results with an analytical study, which shows qualitative agreement with the numerical simulations.

Key words:

1. Introduction

The hydrodynamics of swimmers have been mainly studied in open waters (Lighthill 1971; Triantafyllou et al. 2000) but a lot of swimmers are evolving in complex, confined or cluttered environments (Tokeshi & Arakaki 2012)—such as narrow channels, cavities, or densely structured habitats—where boundaries and obstacles significantly influence their hydrodynamics.

The self-propulsion of a swimmer in a viscous fluid results from the balance between the generated thrust and the drag forces exerted by the surrounding fluid on the immersed body. The relative importance of viscous damping and inertia therefore crucially determines the swimming constitutive law relating the motion velocity to the generated thrust. This can be quantified with the Reynolds number (Re), defined as the non-dimensional ratio between the inertial and viscous forces (see e.g. Kundu et al. (2024)): for low $Re < 1$, viscous diffusion dominates, imposing large momentum dissipation, while for large $Re > 1$, inertia becomes important, leading to a localization of velocity and pressure gradients. Three different universal swimming regimes result from the decreasing impact of viscous forces on the flow as the Reynolds increases: the low Reynolds – or Stokes – regime is the

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realm of micro-organisms (Lauga & Powers 2009; Guasto et al. 2012), for which all the motion is driven by drag forces; the large Reynolds – or turbulent – regime, where inertia controls the flow through momentum transfer, and the intermediate – laminar – regime, where both viscous and inertial effects play a role, which balancing effect is concentrated in a boundary layer close to the immersed body (Anderson et al. 2001).

Moreover, it is already known, for a flow past a cylinder, that the interaction of the boundary layer with walls will increase the critical Reynolds number where the shedding of vortices occurs (Sahin & Owens 2004). It means that the interaction between the wall and the boundary layer is playing a role in the balance between the viscous forces and the inertial effects. This interaction between the boundary layer and the confinement is really important for the optimisation of the swimming behaviour.

Indeed, optimisation strategies in swimming are of major importance among biological organisms. Such optimisation can be observed, for example, in burst-and-coast swimming behaviour (Anderson et al. 2001). The hydrodynamic drag experienced by a rigid body is lower than that experienced by a body undergoing active swimming. The coast phase allows the organism to cruise while minimising drag, following a burst phase that generates thrust. Swimming optimisation can also be observed at the transition between the Stokes and laminar flow regimes. The specie *Clione antartica* is able to change its mode of propulsion according to its Reynolds number (Childress & Dudley 2004). Indeed for $Re < 5$, it uses metachronal waves generated by three rings of cilia to swim. Then, for $Re > 15$, it switches to a flapping-wing propulsion mechanism.

The study of boundary-layer behaviour at intermediate Reynolds numbers, or under confined conditions, remains an open question and could provide insight into swimming optimisation in relation to the environment. In this paper, we investigate numerically the behaviour of a swimmer in a confined environment. We show that the transition from the Stokes regime to the intermediate inertial regime can be driven solely by adjusting the confinement. We further support these results with an analytical study of the boundary layer in a confined environment.

2. Model

To investigate the fundamental role of confinement on the behaviour of a swimmer, we focus exclusively on the forces exerted by the swimmer on the surrounding fluid, utilizing a force-based model, that has been validated across a broad range of Reynolds numbers (Ventéjou et al. 2025). This approach is particularly relevant to our study, as it generates a hydrodynamic flow field characteristic of a force dipole, which corresponds to a pusher-type swimmer at low Reynolds numbers (Lauga & Powers 2009) and a fish-like swimmer at high Reynolds numbers (Ventéjou et al. 2025). Moreover, it provides a way to identify the hydrodynamic regime associated to the swimming by linking the scaling of the Reynolds number to the Thrust number—a non-dimensional number relating the force density, the inertial term and the viscous term (Ventéjou et al. 2025).
As shown in fig. 1, the swimmer is represented by a rigid ellipsoid $\mathcal{B}$ and generates its motion by exerting a force $\boldsymbol{F}_0$ directly in the fluid in $\boldsymbol{X}_t$ by the tail or flagellum. Then, as the system is isolated, we set an opposite force $-\boldsymbol{F}_0$ directly applied to the body $\mathcal{B}$ in $\boldsymbol{X}_{\mathcal{B}}$. The force dipole is described by the distance d and the amplitude F_0. The ellipsoid rigid body is described by a semi-major axis a and a semi-minor axis b. The canal width is defined by the minimal distance h between the wall and the rigid body. Thus, to study

the confinement we use a canal of width $2h + 2b$ and set the body in the middle, so that then the swimmer moves parallel to the wall.

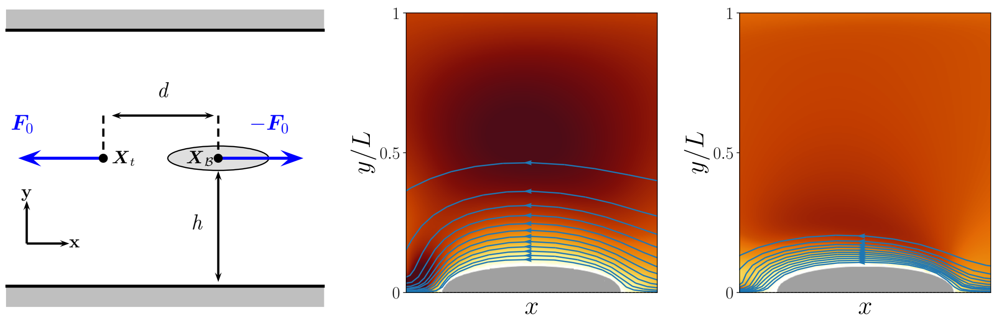

Figure 1. *Left:* Scheme of the model used to describe the swimmer and perform numerical simulations. The swimmer is represented by a rigid body ellipsoid $\mathcal{B}$ located in $\boldsymbol{X}_{\mathcal{B}}$. The force dipole is defined by a couple of force $\{\boldsymbol{F}_0, -\boldsymbol{F}_0\}$ and a distance d. The control parameter representing the canal width is defined by h. The speed of the swimmer is given by U. The upper and bottom limit correspond to the wall with a zero velocity boundary condition. The boundary conditions in the x direction are given by walls with zero velocity boundary conditions and we set the canal length to be much larger $(3 - 50)$ compare to the width h. *Middle:* Streamlines around the swimmer in the Stokes regime with $Re = 7.7$. *Right:* Streamlines around the swimmer in the intermediate inertial regime with $Re = 147$. The grey half ellipsoid corresponds to the swimmer body. Streamlines are plotted in the swimmer frame with a step in velocity $\Delta v_x = 0.1U$. The colormap shows the magnitude of the velocity field.

We perform direct numerical simulations of the incompressible Navier-Stokes equations in the presence of our model swimmer, see fig. 1. The equation reads:

$$\rho\left(\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{v}.\boldsymbol{\nabla})\boldsymbol{v}\right) + \boldsymbol{\nabla} p - \nabla.\left(\eta(\boldsymbol{\nabla v} + \boldsymbol{\nabla v}^T)\right) = \mathcal{F} \tag{2.1}$$

$$\boldsymbol{\nabla}.\boldsymbol{v} = 0 \tag{2.2}$$

where $\boldsymbol{v}$ and p are respectively the velocity and the pressure fields, ρ and η denote the density and the viscosity fields, and $\mathcal{F}$ corresponds to the force field applied by the swimmer. The density field ρ is spatially homogeneous in the whole domain to ensure that the swimmer is neutrally buoyant. The rigid body $\mathcal{B}$ is taken into account with a penalty method where the viscosity is a field defined by $\eta = \eta_f + (\eta_{\mathcal{B}} - \eta_f)H_{\mathcal{B}}$ where $H_{\mathcal{B}}$ is the Heaviside function of $\mathcal{B}$, η_f the viscosity of the fluid, and $\eta_{\mathcal{B}}$ the viscosity inside the rigid body $\mathcal{B}$. To ensure that the rigid body constraint satisfied, we use a ratio $\eta_{\mathcal{B}}/\eta_f = 10^{3-6}$ and check that there is no internal recirculation of fluid inside the body. The force field is defined as $\mathcal{F} = \boldsymbol{F}_0\delta_{\boldsymbol{X}_t} - \boldsymbol{F}_0\delta_{\boldsymbol{X}_{\mathcal{B}}}$ where $\delta_{\boldsymbol{X}}$ is the Dirac distribution centered at $\boldsymbol{X}$. The incompressible Navier-Stokes equations are solved numerically using an implicit $\mathcal{P}2 - \mathcal{P}1$ finite element method (Metivet et al. 2018) implemented in the parallel FEEL++ library (Prud'homme et al. 2012). The position and orientation of the swimmer are updated at each time-step using a first order Euler scheme with the translational and rotational velocities computed from the fluid velocity field in $\mathcal{B}$.

The velocity scale of this problem is given by the velocity U of the swimmer in the stationnary state. The typical length scales of this problem are the size of the swimmer $L = 2a$, the size of the canal width h and the typical size associated to the velocity gradient (Guyon et al. 2015), which is given by the Blasius' definition $\delta^* = L/\sqrt{Re}$. One can notice that the swimmer's configuration is different from the study where the Blasius' definition of the boundary layer is defined— here the force is fixed instead of fixing the velocity of the flow far from the swimmer.

3. Numerical simulations

To study the effect of the confinement on the hydrodynamic regime of the swimmer, we vary two control parameters F_0 and h, while the other parameters are fixed. For a fixed canal width, the Reynolds number Re is set by the force F_0. According to Ventéjou et al. (2025), in the Stokes regime, the Reynolds number scales linearly with the thrust number $Th = \frac{\rho F_0}{\eta^2}$. In the intermediate regime, the Reynolds number scales as $Re \sim Th^{2/3}$. Then we perform numerical simulations for a canal width from $h/L = 0.15$ to $h/L = 5$ and for a force F_0 leading to a Reynolds number from $Re = 8\ 10^{-4}$ to $Re = 250$.
To understand the effect of confinement on the hydrodynamic regime of the swimmer, we first focus on the streamlines around the swimmer, see fig. 1. The plots show around ten streamlines in the swimmer's frame. The middle plot corresponds to a swimmer where the streamlines are spread over half of the width of the canal, it means that the velocity gradient occurs over a typical size which is more or less equal to half of the width. This behaviour corresponds to the Stokes regime according to the linear scaling of the Reynolds number with the thrust number. The right plot corresponds to a swimmer where the streamlines are concentrated close to the swimmer's body. According to the scaling of the Reynolds number with the thrust number, this behaviour corresponds to the intermediate inertial regime. The velocity gradient is localised in a thin layer of fluid. It could be seen as a boundary layer and should play a key role in the swimming behaviour.

4. Boundary layer model in a confined geometry

To understand the role of the boundary layer in controlling the transition between the Stokes and the intermediate inertial swimming regimes, we develop an extension of the Blasius model (Blasius 1907), to account for the presence of a confining wall close to the swimmer.

We consider the idealised configuration of a semi-infinite plate with characteristic dimension L, moving with some constant velocity $-U\boldsymbol{e_x}$ ($U \geq 0$) near a parallel wall (see fig. 2 for a schematic diagram). The distance between the plate and the wall h is supposed much smaller than L. In the moving frame attached to the plate (with origin set at its left end), a boundary layer with characteristic thickness $\delta^* = L/\sqrt{Re}$ develops for $x \geq 0$ to satisfy the no-slip boundary condition at the surface of the plate.

As shown in appendix A, we can obtain a Blasius-like self-similar solution in this confined case using the streamline function ansatz $\psi = \delta(x) f(\eta)$ with $\eta = \frac{y}{\delta(x)}$ into the 2D Navier-Stokes equations. The resulting x-momentum balance equation leads to the familiar "square-root" profile for the boundary layer $\delta(x) = \delta^* \sqrt{x/L}$, along with the differential equation for f:

$$f^{(4)} + \frac{1}{2}(f f''' + f' f'') = 0. \tag{4.1}$$

with the boundary conditions

$$f(0) = f'(0) = 0, \quad f(\eta_h) = \eta_h, \quad f'(\eta_h) = \eta_h \tag{4.2}$$

where $\eta_h = \frac{h}{\delta(x)}$ is defined by the confinement.

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Note that Equation (4.1) can be integrated once to get $f''' + \frac{1}{2} f f'' - \beta = 0$, which closely resembles the standard Blasius' equation, but with an additional constant β representing the unknown – and non-zero – pressure gradient. We also observe that the presence of confinement introduces an x-dependent boundary condition at the wall (i.e. for $\eta = \eta_h(x)$), which makes the integration of the equation more difficult.

The boundary value problem eqs. (4.1) and (4.2) can be solved numerically with an implicit Runge-Kutta method (we used in practice the Lobatto IIIa2 implicit trapezoidal rule implemented in Rackauckas & Nie (2017)), leading to flow velocity profiles as shown in fig. 2.

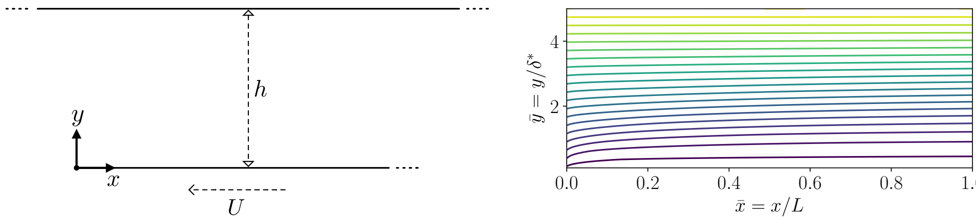

Figure 2. Confined Blasius boundary layer configuration and streamlines. Left: schematic diagram for the semi-infinite plate moving parallel to a confining wall. Right: velocity streamlines in the $(x/L, y/\delta^*)$ plane, computed for a confinement $\bar{h} = h/\delta^* = 5$.

It naturally follows from the self-similarity approach that the solution for the velocity field (in the plate-attached frame) $v_x = Uf'(\eta)$, $v_y = U\delta'(x)(\eta f'(\eta) - f(\eta))$ only depends on the confinement through the non-dimensional combination $\bar{h} = \frac{h}{\delta^*} = \frac{h}{L}\sqrt{Re}$. The effective confinement thus depends on both the geometric confinement, and the swimming Re number.

Figure 3 shows the resulting longitudinal velocity profiles (translated to the lab-frame for comparison with the numerical swimmer data) obtained for different confinements and with the analytical framework and numerical simulations. First, we observe a qualitative agreement between the numerical simulations and the analytical model. In contrast with the usual unconfined Blasius solution (c.f. for example Kundu et al. (2024)), we notice the presence of a backflow, imposed by the fluid mass conservation in a channel. We also observe two regimes: in very confined configuration ($\bar{h} < 5$), the velocity profile is parabolic, with a transverse gradient spread across the channel. This parabolic profile is similar to a Poiseuille flow. While for larger confinements $\bar{h} > 5 - 10$, the velocity gradient localises in a boundary layer, similar to the unconfined Blasius limit.

To quantify the thickness of the boundary layer, we extend the standard "δ_1" displacement thickness (see e.g. Kundu et al. (2024)) to our confined configuration. This thickness corresponds to the width of a virtual zero-velocity fluid leading to the velocity deficit created by the streamlines displacement. In our case, we must however account for the backflow, and therefore only consider the flow deficit in the region $y \in [0; y_0]$ where v_x goes from 0 to U, i.e. y_0 corresponds to the first 0 of the function shown in fig. 3, which can be determined using a standard Newton root-finding algorithm. The boundary layer thickness δ_1 then corresponds to $\int_{y=0}^{y_0} (U - v_x) dy = U\delta_1$, that is

$$\delta_1(x) = \delta^*(\eta_0 - f(\eta_0))\sqrt{\frac{x}{L}} \tag{4.3}$$

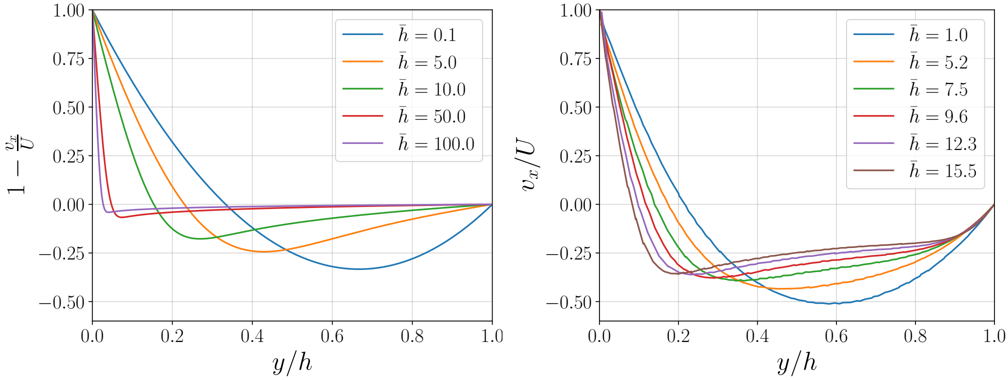

Figure 3. Confined Blasius boundary layer velocity profile. *Left:* Analytical results. Longitudinal velocity in the lab-frame $1-\frac{v_x}{U}$ for different confinements $\bar{h}=\frac{h}{\delta^*}$ ranging from 0.1 to 100. This profile was computed for a vertical slice at $\bar{x}=\frac{x}{L}=0.5$. *Right:* Numerical simulations. Velocity profile of the longitudinal velocity $v_x(y/h)$ divided by the speed of the swimmer U with a canal width h. The Reynolds number Re is varied by changing the force F_0 with a fixed h.

with $\eta_0 = y_0/\delta(x)$. We show in fig. 4 the evolution of the resulting non-dimensional thickness $\bar{\delta}_1 = \delta_1/\delta^*$ obtained from our confined Blasius model for a large range of confinements.

5. Discussion

According to the modelling of the boundary layer, we can extract the value of $\bar{\delta}_1$ from the numerical simulations and from the analytical model. Figure 4 shows the evolution of the boundary layer $\bar{\delta}_1$ as a function of the confinement $\bar{h}$. We plot the data for different Reynolds number values and show a collapse of the numerical data. A good agreement is obtained between the numerical data and the analytical solution. The thickness of the boundary layer $\bar{\delta}_1$ shows two different regimes. For small confinement $\bar{h}$, the boundary layer increases linearly with the confinement $\bar{\delta}_1 \sim \alpha\bar{h}$, with $\alpha \approx 0.15$. A quantitative agreement for the saturation value is not achievable and we only expect a qualitative agreement. Even if the two cases present a similar concept, the implementation presents some differences. Indeed, the numerical simulations are performed with a swimmer with a finite size and thus there exists constraints on the total flow. Moreover, the swimmer is using a force dipole to propel itself and there are differences between a dipole and a monopole, see appendix. Then, for larger confinements $\bar{h}$, the boundary layer saturates to a constant value. In the analytical study, for infinite $\bar{h}$ we recover the same saturation value as the standard Blasius solution (Landau & Lifshitz 1987). The transition between these two regimes is governed by the dashed line $\bar{h} \sim 5$ which, in fact, corresponds to the limit between the Stokes regime and the intermediate inertial regime according to scaling laws of the velocity, with $Re \sim Th$ in the Stokes regime and $Re \sim Th^{2/3}$ in the intermediate regime.

Moreover, it is also possible to recover the scaling laws observed numerically by using the definition of the drag force. The drag is formally given by $F_0 \sim \eta_f \frac{U}{\delta_1} L^2$. In the regime where $\bar{\delta}_1 \sim \bar{h}$, which also reads $\delta_1 \sim h$, after replacing δ_1 in the definition of F_0 it gives $Re \sim Th\frac{h}{L}$. For a fixed canal width and swimmer body, we recover the scaling law observed in the Stokes regime, which is in agreement with what is observed numerically. In the regime where $\bar{\delta}_1 \sim cst$, it leads to $\delta_1 \sim \frac{L}{\sqrt{Re}}$ and we get $Re \sim Th^{2/3}$. This scaling law is also in agreement with what is found numerically for the intermediate regime.

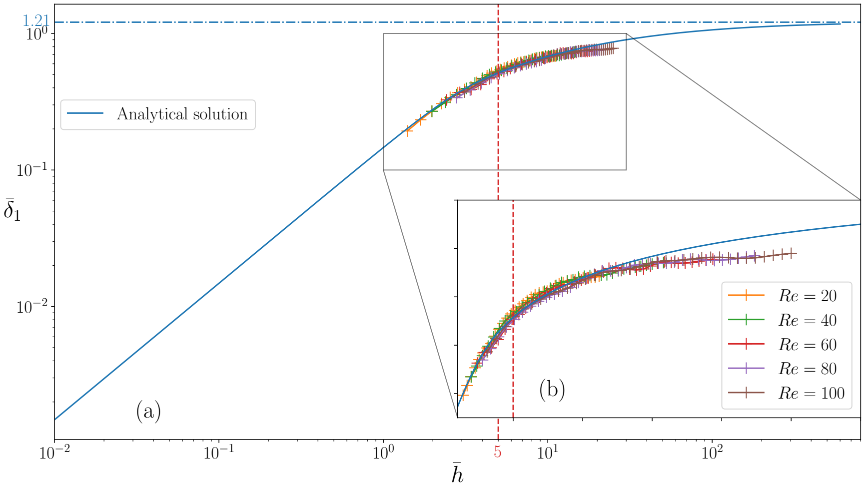

Figure 4. Evolution of the boundary layer $\bar{\delta}_1$ as a function of the confinement $\bar{h}$. (a): Results from analytical solution (blue line) and numerical simulation (colored line with cross) for different Reynolds number values. The dashed line corresponds to the transition between the Stokes and the intermediate regime. The dash dotted line is the saturation value of the analytical solution.

Based on the previous results, we can construct a phase diagram in the plane $(Re, h/L)$. Figure 5 shows the phase diagram where the blue region corresponds to the linear evolution of the boundary layer $\bar{\delta}_1$ as a function of the confinement $\bar{h}$ and the red region shows the saturation of the boundary layer $\bar{\delta}_1$. The dashed red line corresponds to the limit between the Stokes regime and the intermediate inertial regime, given by $\bar{h} = 5$. The grey part corresponds to the Stokes regime whatever is the width of the canal. Indeed, if the force dipole is not strong enough it is never possible to reach the intermediate regime even in open waters because the swimmer is too slow and viscous forces dominate. This phase diagram confirms that it is possible to drive the transition from the Stokes regime to the intermediate regime solely by adjusting the width of the canal h. Indeed, for a given Reynolds number, by increasing h it is possible to switch from Stokes regime to intermediate regime. It also means that it is possible to be in the Stokes regime with $Re \gg 1$ in confined environement. This result shows that knowledge of the Reynolds number alone is not sufficient to understand the hydrodynamic regime of a confined swimmer.

6. Conclusion and perspectives

In this study, we have investigated numerically and analytically the influence of confinement on the hydrodynamic regime of a generic swimmer, independently of its propulsion mechanism. We have shown that the transition from the Stokes regime to the intermediate inertial regime can be controlled solely by tuning the confinement. This transition is governed by the formation of a boundary layer, whose thickness is either imposed by the confining walls or set by inertial effects characterised by the Reynolds number through a Blasius-like scaling. The qualitative agreement between numerical simulations and the confined boundary-layer model highlights the central role of boundary-layer in determining drag scaling and swimming speed. These results demonstrate that the Reynolds number alone is not sufficient to characterise the hydrodynamic regime of a swimmer in confined environments.

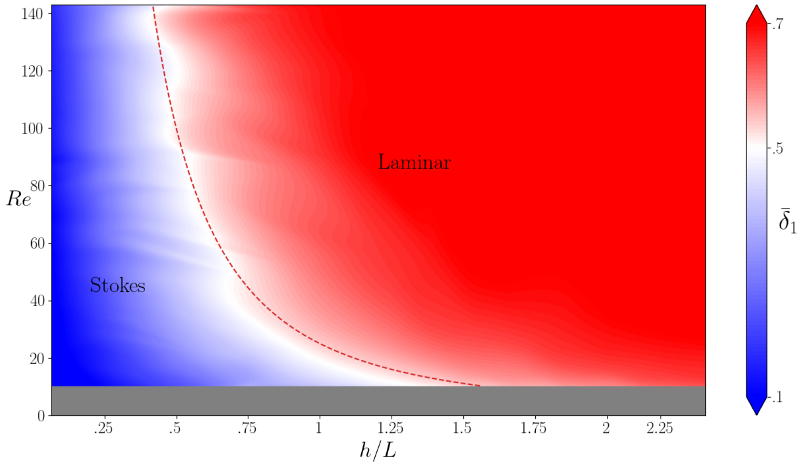

Figure 5. Phase diagram in the plane Reynolds number and canal width $(Re, h/L)$ with L the size of the swimmer. The red part shows the saturation of the boundary layer as a function of the canal width and the blue part shows the Stokes regime. The dashed red line corresponds to the limit between the linear growth of $\bar{\delta}_1$ as a function of $\bar{h}$ and the saturation. It is the transition between the Stokes and the intermediate regime. The dashed red line is given by $\bar{h} = h\sqrt{Re}/L = 5$. The grey area corresponds to the Stokes regime whatever is the width of the canal h.

Several perspectives emerge from this work. Extending the analysis to more complex confinement geometries, such as asymmetric channels or porous media, would provide a more realistic description of natural habitats. It is known that the mode of propulsion through deformation is very different in the Stokes regime or in inertial regime, this means that deformation should be adapted to the different confinements encountered by the swimmer. Finally, studying interactions between multiple swimmers under confinement could shed light on collective effects and on the role of hydrodynamic regimes in biological and artificial swarms.

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Appendix A. Confined Blasius boundary layer solution

Following the usual Blasius scale separation between the longitunal and transverse directions, we introduce the non-dimensional coordinates, velocities and pressure

$$
\begin{aligned}
\bar{x} \equiv \frac{x}{L}, \qquad \bar{y} \equiv \frac{y}{\delta^*} = \frac{y}{L}\sqrt{Re} \\
\bar{v}_x \equiv \frac{v_x}{U}, \quad \bar{v}_y \equiv \frac{v_y}{U}\frac{L}{\delta^*} = \frac{v_y}{U}\sqrt{Re}, \quad \bar{p} \equiv \frac{p}{\rho U^2}.
\end{aligned} \tag{A 1}
$$

The mass and momentum conservation equations then write

$$
\begin{aligned}
\frac{\partial \bar{v}_x}{\partial \bar{x}} + \frac{\partial \bar{v}_y}{\partial \bar{y}} = 0 \\
\bar{v}_x \frac{\partial \bar{v}_x}{\partial \bar{x}} + \bar{v}_y \frac{\partial \bar{v}_x}{\partial \bar{y}} = -\frac{\partial \bar{p}}{\partial \bar{x}} + \frac{1}{Re}\frac{\partial^2 \bar{v}_x}{\partial \bar{x}^2} + \frac{\partial^2 \bar{v}_x}{\partial \bar{y}^2} \\
\frac{1}{Re}\left(\bar{v}_x \frac{\partial \bar{v}_y}{\partial \bar{x}} + \bar{v}_y \frac{\partial \bar{v}_y}{\partial \bar{y}}\right) = -\frac{\partial \bar{p}}{\partial \bar{y}} + \frac{1}{Re^2}\frac{\partial^2 \bar{v}_y}{\partial \bar{x}^2} + \frac{1}{Re}\frac{\partial^2 \bar{v}_y}{\partial \bar{y}^2}
\end{aligned} \tag{A 2}
$$

with boundary conditions

$$\begin{aligned} \bar{v}_x = 0 \text{ and } \bar{v}_y = 0 \quad & \text{at } \bar{y} = 0, \\ \bar{v}_x = 1 \text{ and } \bar{v}_y = 0 \quad & \text{at } \bar{y} = \bar{h}. \end{aligned} \tag{A 3}$$

Assuming that Re is large enough, we can look for a Blasius-like self-similar solution of the leading order equations with the non-dimensional streamline function $\bar{\psi} \equiv \bar{\delta}(\bar{x}) f(\eta)$, where $\eta \equiv \bar{y}/\bar{\delta}(\bar{x})$. The leading order longitudinal momentum conservation equation then lead to

$$\left(\frac{\bar{\delta}'}{\bar{\delta}}\right)[f f''] = \frac{\partial \bar{p}}{\partial \bar{x}} - \frac{f'''}{\bar{\delta}^2}. \tag{A 4}$$

Note that this equation closely resembles the unconfined Blasius boundary layer one, but here the pressure gradient along x does not vanish, as it is required to balance the confining pressure caused by the wall. We can however use the leading order transverse balance $\frac{\partial \bar{p}}{\partial \bar{y}} = 0$, and differentiate eq. (A 4) w.r.t $\bar{y}$, to get

$$\left(\frac{\bar{\delta}'}{\bar{\delta}^2}\right)[f''' f + f'' f'] = -\left(\frac{1}{\bar{\delta}^3}\right)[f^{(4)}]. \tag{A 5}$$

Equating terms which only depend on $\bar{x}$ (resp. η) in (.) (resp. [.]), and enforcing $\bar{\delta} \xrightarrow[\bar{x}\to 0]{} 0$, we finally obtain $\bar{\delta} = \sqrt{\bar{x}}$, and the differential equation eq. (4.1) for f. Note that the we recover Blasius' solution for $\bar{\delta}$, despite the presence of confinement. The thickness profile is however now set by a fourth order ODE, as we need to solve for the pressure gradient in addition to the velocity.

The boundary conditions eq. (4.2) naturally result from eq. (A 3), recalling $\bar{v}_x = f'(\eta)$ and $\bar{v}_y = \bar{\delta}'(\bar{x})(\eta f'(\eta) - f(\eta))$.

Appendix B. Swimmer finite-size effects

In the main text, we have compared the confined Blasius boundary layer model with numerical simulations of a finite-size swimmer. However, the velocity profile obtained from the numerical simulations in fig. 3 are made with a fixed canal width h and we only vary the Reynolds number Re to change the value of the confinement $\bar{h}$. The comparison with analytical model gives a good qualitative agreement. According to this results, one can expect a collaspe, by varying h and Re, of the velocity profile according to the confinement $\bar{h}$. However, this collapse of the velocity profile is not observed, see fig. 6. This observation is due to the finite size of the swimmer. Indeed, the flow need to be conserved in the canal and the swimmer's body contribution depends on the ratio between the width of the canal h and the width of the swimmer b. To expect a collapse of the velocity profile, it is necessary to take into account a correction which depends on the ratio h/b.

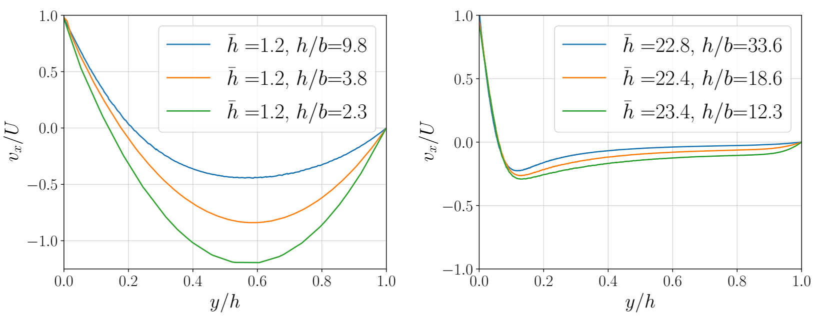

Figure 6. Velocity profile of the longitudinal velocity $v_x(y/h)$ divided by the speed of the swimmer U with a constant confinement $\bar{h}$. *Left:* Stokes regime $\bar{h} < 5$. *Right:* Intermediate regime with $\bar{h} > 5$. The confinement $\bar{h}$ is varied by changing h and the Reynolds number through F_0. The different curves correspond to different values of the ratio h/b where b is the width of the swimmer.

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